# RESEARCH PLAN

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| **Researcher** | Shayan Gul (ORCID: https://orcid.org/0009-0009-1584-7208) | **Supervisor** | Prof. Reyes Mallada |
| **Host** | NFP group, INMA, University of Zaragoza | **Doctoral programme** | Chemical Engineering |
| **Research line** | $CO_2$ mitigation: capture, utilisation and process decarbonisation | **Period** | 48 months, started from 19 Feb 2026 |

## 1. Summary

Regenerating solid $CO_2$ adsorbents by conventional heating is slow and energy-intensive, because heat must travel through the bed from the walls. This project studies **Microwave Swing Adsorption (MWSA)**, in which microwaves heat the adsorbent directly and volumetrically, combined with **3D-printed structured adsorbents** made by direct ink writing (DIW), whose geometry can be designed for both gas–solid contact and microwave heating. **WP1** develops and tests zeolite-based monoliths for $CO_2$ capture with microwave regeneration, supported to scale-up. **WP2** extends the concept to $CO_2$ utilisation through **microwave-assisted dry reforming of methane (DRM)** in structured reactors. The project links chemical engineering at UNIZAR with microwave engineering at UPV and includes secondments at VITO and Ghent University (Belgium).

**Keywords:** $CO_2$ capture; microwave swing adsorption; direct ink writing; structured adsorbents; process intensification; dry reforming of methane.

## 2. Hypotheses and objectives

- **H1.** Monolith geometry and infill can be designed to improve both $CO_2$ adsorption and the uniformity of microwave heating.
- **H2.** Microwave regeneration of printed zeolite monoliths is faster than conventional heating while keeping the capacity stable over cycles.

**General objective:** to develop 3D-printed structured adsorbents that are efficiently regenerated by microwaves for $CO_2$ capture, and to extend the concept to $CO_2$ utilisation, contributing to electrified and intensified $CO_2$ mitigation processes.

| No. | Specific objective | Task | Indicator of success |
|---|---|---|---|
| **O1** | Formulate printable zeolite pastes and print monoliths of different geometries and infill patterns | T1.1 | Reproducible monoliths (copies a, b, c) verified by XCT. |
| **O2** | Evaluate $CO_2$ adsorption and microwave regeneration, including temperature distribution | T1.2 | $CO_2$ Ads des Capacity, regeneration performance and IR temperature profiles |
| **O3** | Model to scale up the structured adsorption system | T1.3 | Scaled-up prototype |
| **O4** | Develop catalysts and structured reactors for microwave-assisted DRM and optimise the process | T2.1–T2.3 | Conversion and process performance data |

## 3. Methodology and work packages

The work follows an iterative design–build–test cycle: paste formulation and rheology, design (Solid Edge) and slicing (Cura, Sygnis), DIW printing, calcination, characterisation, and microwave adsorption–desorption testing, with the results feeding back into the next monolith design. Each design is printed in three copies (a, b, c) to assess repeatability; the first systematic set (batch B6, zeolite, zigzag geometry) covers four infill patterns (10, 18, 25 and 35), giving twelve monoliths.

| Task | Months | Methods | Expected output |
|---|---|---|---|
| **T1.1** 3D printing and characterisation | M1–M24 | Rheology, DIW, calcination, TGA, XCT | Optimised pastes and a library of monoliths |
| **T1.2** Microwave regeneration performance | M1–M24 | MW adsorption–desorption cycles, mass spectrometry, IR camera; dielectric behaviour with the help UPV team. | Performance and cycle stability versus design |
| **T1.3** Scale-up of the structured system | M13–M36 | Selection of the best design; larger structured system | Scaled-up system and assessment |
| **T2.1** Materials and catalysts | M13–M36 | Catalysts and supports suited to MW heating | DRM catalysts |
| **T2.2** MW-assisted DRM in structured reactors | M13–M36 | Structured catalytic reactors under microwave heating | Conversion and product data |

| Task | Months | Methods | Expected output |
|---|---|---|---|
| **T2.3** Process optimisation | M13–M48 | Optimisation of conditions; performance and energy assessment | Optimised process |

## 4. Work plan and milestones

| Activity (M1 = Feb 2026) | Y1 (2026–27) | | | | Y2 (2027–28) | | | | Y3 (2028–29) | | | | Y4 (2029–30) | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Quarter** | Q1 | Q2 | Q3▼ | Q4 | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 | Q3 | Q4 | Q1 | Q2 | Q3 | Q4 |
| **WP1 $CO_2$ structured adsorbents regenerated by microwaves** | | | | | | | | | | | | | | | | |
| T1.1 3D printing and characterisation | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | | | | | | | | |
| T1.2 Microwave regeneration performance | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | | | | | | | | |
| T1.3 Scale-up of the structured system | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | | | | | | | | |
| **WP2 Microwave-assisted dry reforming of methane** | | | | | | | | | | | | | | | | |
| T2.1 Materials and catalysts | | | | | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | | | | |
| T2.2 MW-assisted DRM in structured reactors | | | | | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | | | | |
| T2.3 Process optimisation and assessment | | | | | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ |
| **Mobility** | | | | | | | | | | | | | | | | |
| Short stays: UPV (Nov 2026), CEINMAT (Feb 2029) | | | | ■ | | | | | | | | | ■ | | | |
| Secondments: VITO (2027), Ghent University (2028) | | | | | ■ | ■ | | | | ■ | | | | | | |
| **Thesis and dissemination** | | | | | | | | | | | | | | | | |
| Publications and conferences | | | | | | | ■ | | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ |
| Thesis writing and defence | | | | | | | | | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ |

*Table: four-year work plan by quarter. ▼ marks the current quarter (M8, September 2026).*

## 5. Progress to date (M1–M8)

| Area | Achievement | Status |
|---|---|---|
| **Literature review** | Systematic review of $CO_2$ capture technologies, focused on MWSA, 3D-printed adsorbents and microwave regeneration | Done, updated |
| **Printing and materials** | DIW skills; Zeolite based monoliths of different geometries designed, printed and calcined. | Done |
| **Characterisation** | Rheometry and TGA of the ink, and XCT of printed materials and monoliths | Ongoing |
| **Sample set B6** | Zeolite, zigzag geometry, infill 10/18/25/35, three copies each (12 monoliths) | Done |
| **Training** | **Ethics and Research Integrity** (4 h, 1 Jul 2026) | Completed |
| **Training** | **Open Science and Data Management** (4 h, 10 Sep 2026). | Completed |
| **Conference** | **IV CEDIC 2026**, Congress of Doctoral Students of the Campus Iberus universities, UPNA, Pamplona, May 2026 | Attended |
| **Management and communication** | Weekly meetings with Prof. Reyes Mallada, jointly with Dr. José Antonio Albajez and Dr. Lucía Díaz (UNIZAR, 3D printing and materials chemistry); weekly NFP group meetings; monthly online meetings with Prof. Felipe Peñaranda (UPV, DIMAS group). | Ongoing |

## 6. Training plan

| Training | Type | Year | Status |
|---|---|---|---|
| Ethics and Research Integrity (4 h) | Mandatory | 2026 | **Completed** |
| Open Science and Data Management (4 h) | Mandatory | 2026 | **Completed** |

| Training | Type | Year | Status |
|---|---|---|---|
| Gender and Diversity in Research (4 h) | Mandatory | 2027 | Planned |
| Developing a Mental Well-being Action Plan (4 h) | Mandatory | 2028 | Planned |
| Research transfer to industry and policy-makers (3 days) | Mandatory | 2029 | Planned |
| Campus Iberus Doctoral Days (3 days, every June) | Mandatory event | Annual | Planned |
| Ethics of AI (2 h); digital skills (2 h); negotiation (2 h); Writing Retreat (3 days); career webinars (8 h) | Optional | 2026–2029 | Planned |
| Discipline-specific and transferable training (120 h): process intensification 25 h; modelling and data analysis 25 h; experimental design 20 h; project management and dissemination 20 h; AI and digital tools 15 h; writing and publication 15 h | Competence development | 2026–2030 | Planned |

## 7. Mobility and the MSCA 3i dimensions

| Host | 3i dimension | Dates | Purpose |
|---|---|---|---|
| UPV, ITACA Institute, DIMAS group (Valencia) | Interdisciplinary | 2 weeks, Nov 2026 | Microwave engineering and dielectric behaviour [Planned] |
| VITO, research institute (Mol, Belgium) | Intersectoral, international | 3 months, Mar–May 2027 | Mandatory non-academic secondment [Planned] |
| Ghent University, Laboratory for Chemical Technology, Prof. Kevin M. Van Geem | International | 3 months, May–Jul 2028 | Thermochemical reaction engineering for WP2 [Planned] |
| CEINMAT, company (Valencia) | Intersectoral | 2 weeks, Feb 2029 | [Planned] |

**Supervision and collaboration:** weekly meetings with Prof. Reyes Mallada, jointly with Dr. José Antonio Albajez and Dr. Lucía Díaz (UNIZAR, 3D printing and materials chemistry); weekly NFP group meetings; monthly online meetings with Prof. Felipe Peñaranda (UPV, DIMAS group).

## 8. Dissemination, exploitation and open science

| Planned event | Location | Date | Participation |
|---|---|---|---|
| EPIC 2027, European Process Intensification Conference | Bologna, Italy | Sep 2027 | Oral or poster |
| ECCE16 & ECAB9 2027 | Edinburgh, UK | Oct 2027 | Oral or poster |
| Leuven Additive Manufacturing Conference 2027 | Leuven, Belgium | To be confirmed | Speaker or attendee |
| AMPERE; UCRA; International and European Microwave Week | To be confirmed | 2028–2029 | Candidate events |

- **Publications:** three to five articles in gold open-access journals (chemical engineering, materials science, process intensification).
- **Communication and exploitation:** “PhD thesis in 3 minutes”; protection of patentable formulations before publication (UNIZAR and Iberus INTER-DOC rules); Iberus–Exolum “Emprende” awards; master’s teaching in English (max. 60 h per year).
- **Open science and ethics:** data are managed under the separate DMP v1.1 (Zenodo with DOI, Zaguán for articles and thesis, “as open as possible, as closed as necessary”). No human participants, animals or personal data are involved.

## 9. Expected impact

- **Career:** an independent researcher with international, intersectoral and interdisciplinary experience, ready for a competitive postdoctoral position in academia or industry.

## 10. Next steps

**Next six months (Oct 2026 to Mar 2027):** finish characterising batch B6 and start microwave tests; short stay at UPV (November 2026); first datasets on Zenodo and upload data on shared OneDrive; complete the Gender and Diversity training; update this plan and the DMP in March 2027.